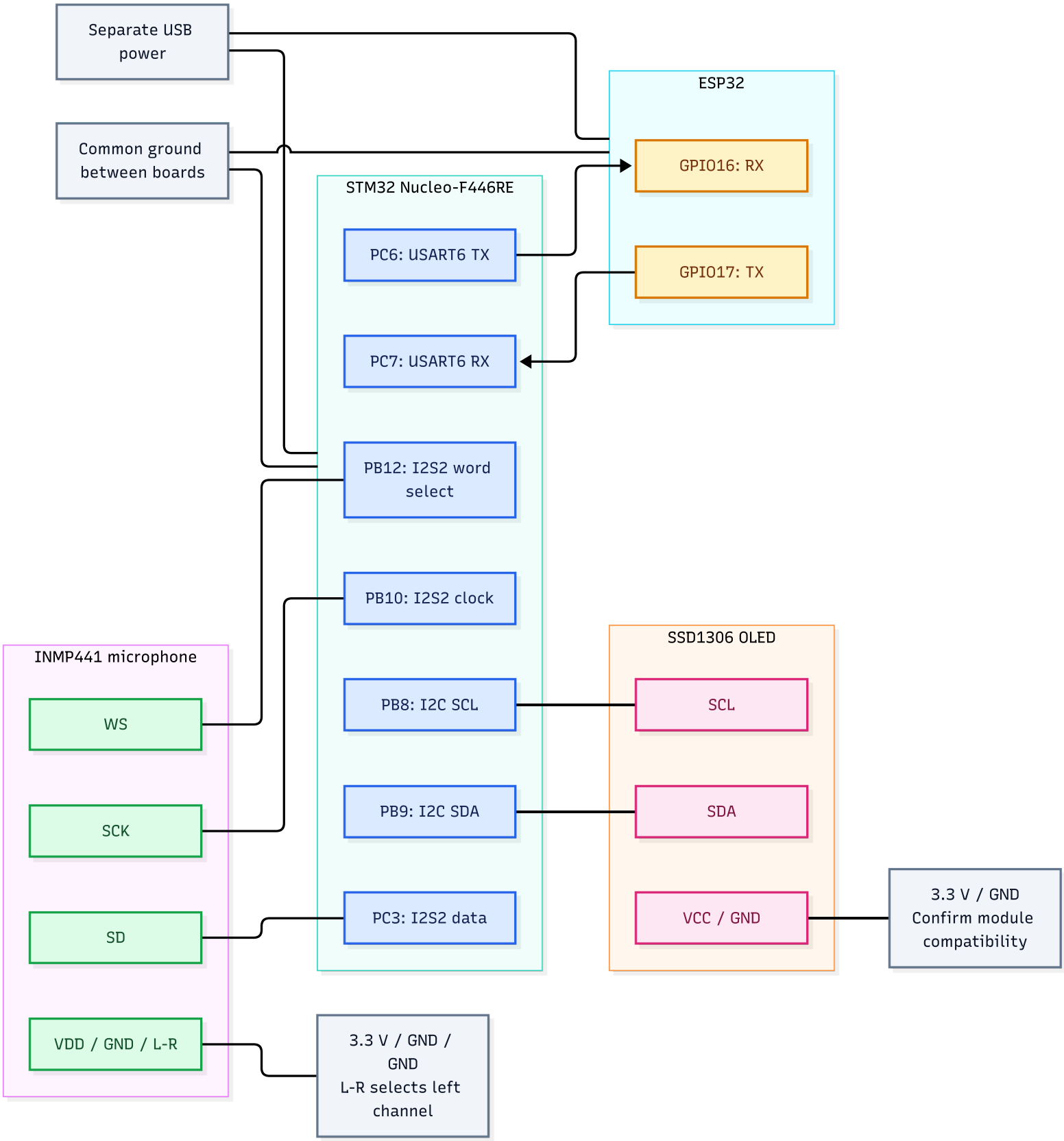

Separate USB power
Common ground between boards
ESP32
GPIO16: RX
GPIO17: TX
STM32 Nucleo-F446RE
PC6: USART6 TX
PC7: USART6 RX
PB12: I2S2 word select
PB10: I2S2 clock
PB8: I2C SCL
PB9: I2C SDA
PC3: I2S2 data
INMP441 microphone
WS
SCK
SD
VDD / GND / L-R
3.3 V / GND / GND
L-R selects left channel
SSD1306 OLED
SCL
SDA
VCC / GND
3.3 V / GND
Confirm module compatibility